Chapter 7

Medical Bacteriology

Learning Objectives

After studying this chapter the students will be able to,

- *Understand the importance of Medical bacteriology.*
- *Describe the pathogenesis of various bacterial diseases such as skin, respiratory infection, food poisoning, diarrhea and dysentery diseases, STD and zoonotic diseases.*
- *Know the collection of appropriate clinical specimens and laboratory diagnosis of various bacterial infections.*
- *Specify how bacterial infections are prevented and treated with appropriate antibiotics.*

Chapter Outline

7.1 Pathogenic Attributes

7.2 Routes of Entry

7.3 *Staphylococcus Aureus*

7.4 *Streptococcus Pyogenes*

7.5 *Neisseria Meningitidis*

7.6 *Corynebacterium Diphtheriae*

7.7 *Clostridium Tetani*

7.8 *Shigella Dysenteriae*

7.9 *Salmonella Typhi*

7.10 *Vibrio Cholerae*

7.11 *Mycobacterium Tuberculosis*

7.12 *Treponema Pallidum*

7.13 *Leptospira Interrogans*

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Medical Bacteriology is the subset of Medical microbiology, which deals with the study of bacterial pathogens. It includes the pathogenesis, diagnosis, treatment and prevention of various bacterial diseases. **Robert Koch** is considered as the **Father of Bacteriology.**

7.1 Pathogenic Attributes

The host-parasite relationship is determined by the interaction between host factors and the infecting pathogens. Pathogenicity refers to the ability of a pathogen to produce disease. Virulence is the ability of the pathogen to cause disease.

Adhesion, invasiveness (Streptococcal infections), Bacterial toxins (endotoxins and exotoxins), capsule enzymes (proteases, collagenase, coagulase and other enzymes). These are already explained in the XI Standard text book.

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